US Patent & Trademark Office
Patent Public Search | Text View

United States Patent 12402331
Kind Code B2
Date of Patent August 26, 2025
Inventor(s) Park; Changyok

Decoupling capacitors based on dummy through-silicon-via plates

Abstract

Disclosed herein are IC structures with decoupling capacitors based on dummy TSV plates provided in a support structure (e.g., a substrate, a die, a wafer, or a chip). An example decoupling capacitor includes first and second capacitor plates and a capacitor insulator between them. Each capacitor plate is a different blind, plate-like opening in the support structure, the openings at least partially filled with one or more conductive materials. The capacitor plate openings are referred to herein as “dummy TSV plates” because they may be fabricated while providing regular TSV openings in the support structure. Such decoupling capacitors may be better suited for high-speed microprocessor applications than conventional off-chip decoupling capacitors and may advantageously allow integrating on-chip decoupling capacitors with an ample amount of capacitive decoupling, limited or no additional processing steps on top of regular TSV processing, and in areas that may not have been used otherwise.

Inventors: **Park; Changyok (Portland, OR)**

Applicant: **Intel Corporation** (Santa Clara, CA)

Family ID: **1000008782575**

Assignee: **Intel Corporation (Santa Clara, CA)**

Appl. No.: **17/170951**

Filed: **February 09, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20220254872 A1	Aug. 11, 2022

Publication Classification

Int. Cl.: **H10D1/00** (20250101); **H01G4/012** (20060101); **H01G4/30** (20060101); **H01G4/38** (20060101); **H01L21/768** (20060101); **H01L23/48** (20060101); **H10D1/68** (20250101)

U.S. Cl.:

CPC **H10D1/042** (20250101); **H01G4/012** (20130101); **H01G4/30** (20130101); **H01G4/38** (20130101); **H01L21/76898** (20130101); **H01L23/481** (20130101); **H10D1/716** (20250101);

Field of Classification Search

CPC: H01L (28/90-92); H01L (29/66181); H01L (29/945); H01L (21/76898); H01G (4/012); H01G (4/30-302); H01G (4/38)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5290729	12/1993	Hayashide	257/E21.018	H01L 28/82
9385039	12/2015	Sakuma	N/A	N/A
9536784	12/2016	Farooq et al.	N/A	N/A
9847255	12/2016	Lin et al.	N/A	N/A
10181454	12/2018	Park	N/A	N/A
2010/0052099	12/2009	Chang	257/E29.342	H01L 23/642
2012/0168902	12/2011	Zhu et al.	N/A	N/A
2015/0028450	12/2014	Park et al.	N/A	N/A
2015/0102395	12/2014	Park et al.	N/A	N/A
2017/0110420	12/2016	Cheng	N/A	H01L 21/6835
2018/0158897	12/2017	Gong	N/A	H01L 28/87
2020/0381420	12/2019	Chen	N/A	H01L 21/823481
2021/0296283	12/2020	Huang	N/A	H01L 23/481
2022/0028825	12/2021	Jeng	N/A	H01L 25/50

OTHER PUBLICATIONS

Extended European Search Report in European Patent Application No. EP22150099 dated Jun. 2, 2022. 11 pages. cited by applicant

Wolf, Stanley, Chapter 16 : Copper Interconnect Process Technology : Silicon Processing for the VLSI Era, vol. 4: Deep-Submicron Process Technology, pp. 711-793, Jan. 2022. cited by applicant

---

Primary Examiner: Kim; Christine S.

Assistant Examiner: Wiegand; Tyler J

Attorney, Agent or Firm: Akona IP PC

Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to the field of integrated circuit (IC) structures and devices, and more specifically, to decoupling capacitors monolithically integrated in such IC structures and devices.

BACKGROUND

(2) A decoupling capacitor is a capacitor used to decouple one part of an electrical network from another. Noise caused by other circuit elements can be shunted through the decoupling capacitor, reducing the effect it has on the rest of the circuit.

(3) Decoupling capacitors are typically included in semiconductor packages in order to lower the inductance through the package by reducing the lead length. Decoupling capacitors placed close to power consuming circuits are able to smooth out voltage variation with charges stored on them. The stored charge either dissipates or is used as a local power supply to device inputs during signal switching stages, allowing the decoupling capacitors to negate the effects of voltage noise induced into the system by parasitic inductance.

(4) Conventional off-chip decoupling capacitors may not be suitable for very high-speed microprocessor applications. Since off-chip decoupling capacitors are located at a relatively long distance from the switching circuits, the time delay caused by the long inductance path makes such capacitors unusable with gigahertz switching circuits.

(5) In order to sustain high frequency circuit operation, an ample amount of capacitive decoupling must be provided close to the switching circuits. Although it is possible to integrate chip capacitors within the chip's circuit elements, the capacitors compete for valuable die area that could be used for building additional circuits. Due to the limited area in which to build these capacitors, the overall capacitive decoupling that they provide is also limited.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

(2) FIG. **1** is a flow diagram of an example method for fabricating decoupling capacitors based on dummy through-silicon-via (TSV) plates, in accordance with some embodiments.

(3) FIGS. **2**A-**2**J illustrate various stages in the manufacture of an example IC structure according to the method of FIG. **1**, in accordance with some embodiments.

(4) FIGS. **3**A and **3**B are top views of a wafer and dies that include one or more IC structures with decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein.

(5) FIG. **4** is a cross-sectional side view of an IC device that may include one or more IC structures with decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein.

(6) FIG. **5** is a cross-sectional side view of an IC device assembly that may include one or more IC structures with decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein.

(7) FIG. **6** is a block diagram of an example computing device that may include one or more IC structures with decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

Overview

(8) The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for all desirable attributes disclosed herein. Details of one or more implementations of the subject matter described in this specification are set forth in the description below and the accompanying drawings.
(9) For purposes of illustrating decoupling capacitors based on dummy TSV plates, described herein, it might be useful to first understand phenomena that may come into play during IC fabrication. The following foundational information may be viewed as a basis from which the present disclosure may be properly explained. Such information is offered for purposes of explanation only and, accordingly, should not be construed in any way to limit the broad scope of the present disclosure and its potential applications.
(10) ICs commonly include electrically conductive microelectronic structures, which are known in the arts as vias, to electrically connect metal lines or other interconnects above the vias to metal lines or other interconnects below the vias. In this context, the term "metallization stack" may be used to describe a stacked series of electrically insulated metallic interconnecting wires that are used to connect together various devices of an IC, where adjacent layers of the stack are linked together through the use of electrical contacts and vias.
(11) Vias are typically formed by a lithographic process. Representatively, a photoresist layer may be spin coated over a dielectric layer, the photoresist layer may be exposed to patterned actinic radiation through a patterned mask or reflected from a patterned mask (e.g., in the case of EUV), and then the exposed layer may be developed in order to form an opening in the photoresist layer, which may be referred to as a via location opening. Next, an opening for the via may be etched in the dielectric layer by using the location opening in the photoresist layer as an etch mask. This opening in the dielectric layer is referred to as a via opening. Finally, the via opening may be filled with one or more metals or other conductive materials to form the via.
(12) TSVs are a particular type of vias that extend through a support structure (e.g., through a substrate, wafer, or a chip). Such regular TSVs are typically used for provision of power and/or signals of various components of IC structures/devices.
(13) Disclosed herein are IC structures (e.g., IC devices) with one or more decoupling capacitors based on dummy TSV plates provided in a support structure (e.g., a substrate, a die, a wafer, or a chip) of an IC structure. An example decoupling capacitor includes first and second capacitor plates and a capacitor insulator between them. Each capacitor plate is a different blind opening in the support structure, the openings at least partially filled with one or more electrically conductive materials. The capacitor plate openings are plate-like openings in that each capacitor plate opening has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth. Thus, as used herein, an opening is described as a "plate-like" if the width of the opening (e.g., a dimension measured along the x-axis of the x-y-z coordinate system shown in the present drawings) is smaller than each of the length of the opening (e.g., a dimension measured along the y-axis of the x-y-z coordinate system shown in the present drawings) and the depth of the opening (e.g., a dimension measured along the z-axis of the x-y-z coordinate system shown in the present drawings), e.g., at least about 2 times smaller, e.g., at least about 4 times smaller or at least about 6 times smaller. In various embodiments, the length of the opening may be either comparable than the depth of the opening or may be larger or smaller than the depth of the opening. Because such plate-like openings may be fabricated substantially simultaneously with providing TSV openings in the support structure, while not being used for the same purposes as the TSVs but rather being used as basis for forming capacitor plates of one or more decoupling capacitors monolithically integrated in the support structure, they are referred to herein as "dummy TSV plates." Such decoupling capacitors may, advantageously, be provided on the same support structures as (e.g., monolithically integrated with) the TSVs and various IC components, making these decoupling capacitors better suited for high-speed microprocessor applications compared to conventional off-chip decoupling capacitors. Furthermore, providing decoupling capacitors in the

support structures of IC structures with capacitor plates being based on dummy TSV plates advantageously allows integrating on-chip decoupling capacitors with an ample amount of capacitive decoupling, limited amount of or no additional processing steps on top of regular TSV processing, in areas that may not have been used otherwise. Because the material of the support structure/substrate (e.g., silicon) may have a higher dielectric constant (e.g., silicon has a dielectric constant of around 12), it may advantageously help with more polarization to store more charge (i.e., higher capacitance of such decoupling capacitors), once electrical isolation is ensured with insulators.

(14) IC structures as described herein, in particular IC structures with TSVs and with decoupling capacitors based on dummy TSV plates as described herein, may be used for providing electrical connectivity to one or more components associated with an IC or/and between various such components. In various embodiments, components associated with an IC include, for example, transistors, diodes, power sources, resistors, capacitors, inductors, sensors, transceivers, receivers, antennas, etc. Components associated with an IC may include those that are mounted on IC or those connected to an IC. The IC may be either analog or digital and may be used in a number of applications, such as microprocessors, optoelectronics, logic blocks, audio amplifiers, etc., depending on the components associated with the IC. The IC may be employed as part of a chipset for executing one or more related functions in a computer.

(15) For purposes of explanation, specific numbers, materials and configurations are set forth in order to provide a thorough understanding of the illustrative implementations. However, it will be apparent to one skilled in the art that the present disclosure may be practiced without the specific details or/and that the present disclosure may be practiced with only some of the described aspects. In other instances, well-known features are omitted or simplified in order not to obscure the illustrative implementations.

(16) Further, references are made to the accompanying drawings that form a part hereof, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense. For convenience, if a collection of drawings designated with different letters are present, e.g., FIGS. **2**A-**2**J, such a collection may be referred to herein without the letters, e.g., as “FIG. **2**.”

(17) In the drawings, some schematic illustrations of example structures of various devices and assemblies described herein may be shown with precise right angles and straight lines, but it is to be understood that such schematic illustrations may not reflect real-life process limitations which may cause the features to not look so “ideal” when any of the structures described herein are examined using e.g., scanning electron microscopy (SEM) images or transmission electron microscope (TEM) images. In such images of real structures, possible processing defects could also be visible, e.g., not-perfectly straight edges of materials, tapered vias or other openings, inadvertent rounding of corners or variations in thicknesses of different material layers, occasional screw, edge, or combination dislocations within the crystalline region, and/or occasional dislocation defects of single atoms or clusters of atoms. There may be other defects not listed here but that are common within the field of device fabrication.

(18) Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

(19) For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and

B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The term “between,” when used with reference to measurement ranges, is inclusive of the ends of the measurement ranges.

(20) The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. The terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. The disclosure may use perspective-based descriptions such as “above,” “below,” “top,” “bottom,” and “side”; such descriptions are used to facilitate the discussion and are not intended to restrict the application of disclosed embodiments. The accompanying drawings are not necessarily drawn to scale. Unless otherwise specified, the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

(21) In the following detailed description, various aspects of the illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art. For example, as used herein, a “high-k dielectric” refers to a material having a higher dielectric constant than silicon oxide while a “low-k dielectric” refers to a material having a lower dielectric constant than silicon oxide. The terms “substantially,” “close,” “approximately,” “near,” and “about,” generally refer to being within +/−20% of a target value based on the context of a particular value as described herein or as known in the art.

(22) Fabricating IC Structures with Decoupling Capacitors Based on Dummy TSV Plates

(23) FIG. **1** is a flow diagram of an example method **100** for fabricating decoupling capacitors based on dummy TSV plates, in accordance with some embodiments. FIGS. **2**A-**2**J illustrate top-down and cross-sectional side views for various stages in the manufacture of an example IC structure **200** (e.g., the IC structure **200**A shown in FIG. **2**A, **200**B shown in FIG. **2**B, and so on until **200**J shown in FIG. **2**J) according to the fabrication method **100**, in accordance with some embodiments. In particular, each of FIGS. **2**A-**2**E shows, at the top of the page of the drawing, a top-down view (i.e., a view in a x-y plane) of the IC structure **200** and, at the bottom of the page of the drawing, a cross-section side view of the IC structure **200** with the cross-section taken along an x-z plane AA′ of the reference coordinate system x-y-z shown in FIGS. **2**A-**2**E. FIG. **2**F shows only the top-down view of the IC structure **200**, FIGS. **2**G and **2**I show only the cross-section side views of the IC structure **200** with the cross-section taken along the x-z plane AA′ indicated in the top-down view of FIG. **2**F, while FIGS. **2**H and **2**J show only the cross-section side views of the IC structure **200** with the cross-section taken along the x-z plane BB′ indicated in the top-down view of FIG. **2**F.

(24) A number of elements referred to in the description of FIGS. **2**A-**2**J with reference numerals are illustrated in these figures with different patterns, with a legend showing the correspondence between the reference numerals and patterns being provided at the bottom of each drawing page containing FIGS. **2**A-**2**J. For example, the legend illustrates that FIGS. **2**A-**2**J use different patterns to show a support structure **202**, a dielectric material **204**, an electrically conductive material **208**, etc. Furthermore, although a certain number of a given element may be illustrated in some of FIGS. **2**A-**2**J (e.g., 4 TSVs and 5 dummy TSV plates in between), this is simply for ease of illustration, and more, or less, than that number may be included in an IC structure according to various embodiments of the present disclosure. Still further, various IC structure views shown in FIGS. **2**A-**2**J are intended to show relative arrangements of various elements therein, and that various IC structures, or portions thereof, may include other elements or components that are not illustrated (e.g., transistor portions, various components that may be in electrical contact with any of the TSVs, etc.).

(25) Turning to FIG. **1**, the method **100** may begin with a process **102** that includes providing openings for one or more TSVs and one or more dummy TSV plates in a support structure. An IC

structure **200**A, depicted in FIG. **2**A, illustrates an example result of the process **102**. As shown in FIG. **2**A, the IC structure **200**A may include a support structure **202**, openings **220-1** through **220-4** for TSVs, and openings **222-1** through **222-5** for dummy TSV plates. Although FIG. **2**A and subsequent FIGS. **2**B-**2**J illustrate an embodiment of the IC structure **200** with 4 TSV openings **220** and 5 dummy TSV plate openings **222**, this is simply for ease of illustration, and more, or less, than that number may be included in the IC structure **200** according to various embodiments of the present disclosure, as long as at least two dummy TSV plate openings **222** (e.g., the dummy TSV plate openings **222-1** and **222-2**) are provided to form first and second capacitor plates of a single decoupling capacitor (presence of the TSV openings **220** is optional).

(26) In general, implementations of the disclosure may be formed or carried out on a substrate, such as a semiconductor substrate composed of semiconductor material systems including, for example, N-type or P-type materials systems. In one implementation, the semiconductor substrate may be a crystalline substrate formed using a bulk silicon or a silicon-on-insulator substructure. In other implementations, the semiconductor substrate may be formed using alternate materials, which may or may not be combined with silicon, that include but are not limited to germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, indium gallium arsenide, gallium antimonide, or other combinations of group III-V, group II-VI, or group IV materials. Although a few examples of materials from which the substrate may be formed are described here, any material that may serve as a foundation upon which an IC may be built falls within the spirit and scope of the present disclosure. In various embodiments, the support structure **202** may include any such substrate, possibly with some layers and/or devices already formed thereon, not specifically shown in the present figures, providing a suitable surface for forming the decoupling capacitors based on dummy TSV plates, as described therein.

(27) Although not specifically shown in the present drawings, any suitable processes may be used in the process **102** to form the TSV openings **220** and the dummy TSV plate openings **222**, e.g., any suitable lithographic process in combination with a suitable etching process. In various embodiments, suitable lithographic processes may include photolithography, electron-beam lithography, etc., which may be used to define locations and dimensions for the TSV openings **220** and the dummy TSV plate openings **222**. In various embodiments, suitable etching processes may include dry etch, wet etch, etc., which may be used to remove portions of the support structure **202** in regions defined by the lithographic process to form the TSV openings **220** and the dummy TSV plate openings **222**. For example, any suitable anisotropic etch process, e.g., a dry etch, may be used in the process **102** to etch the support structure **202** through the openings defined by the lithographic process (e.g., openings defined in a photoresist material, not shown in FIG. **2**A) to form the TSV openings **220** and the dummy TSV plate openings **222**. In some embodiments, during the etch of the support structure **202** in the process **102**, the IC structure may be heated to elevated temperatures, e.g., to temperatures between about room temperature and 200 degrees Celsius, including all values and ranges therein, to promote that byproducts of the etch are made sufficiently volatile to be removed from the surface.

(28) In some embodiments, the width (a dimension measured along the x-axis of the example coordinate system x-y-z shown in FIG. **2**) of the TSV openings **220** may be between about 100 nanometers and 20 micrometers (in various embodiments, TSV opening sizes could hundreds of nanometers to tens of micrometers), including all values and ranges therein, e.g., between about 500 nanometers and 10 micrometers. In some embodiments, the width of the dummy TSV plate openings **222** may be smaller than the width of the TSV openings **220** (e.g., to use micro-loading effect and minimize additional process steps), e.g., the width of the dummy TSV plate openings **222** may be between about 250 and 5000 nanometers, including all values and ranges therein, e.g., between about 100 and 2500 nanometers. In some embodiments, the width of the TSV openings **220** may be between about 1.5 and 10 times greater than the width of the dummy TSV plate openings **222**, including all values and ranges therein, e.g., between about 2 and 5 times greater.

Because the width of the TSV openings **220** is larger than that of the dummy TSV plate openings **222**, the TSV openings **220** may be etched further/deeper into the support structure **202** than the dummy TSV plate openings **222**. In some embodiments, the depth (a dimension measured along the z-axis of the example coordinate system shown in FIG. **2**) of the TSV openings **220** may be between about 1.1 and 4 times greater than the depth of the dummy TSV plate openings **222**, including all values and ranges therein, e.g., between about 1.5 and 2 times greater. In some embodiments, the depth of the TSV openings **220** may be between about 1.1 and 4 times smaller than the thickness of the support structure **202** (i.e., a distance between a first side **224-1** and a second side **224-2** of the support structure **202**, a dimension measured along the z-axis of the example coordinate system shown in FIG. **2**). In some embodiments, the depth of the TSV openings **220** may be between about 500 nanometers and 50 micrometers (it could be hundreds of nanometers to tens of micrometers, for regular TSVs it could be few to tens of micrometers, e.g., 2 to 50 micrometers), including all values and ranges therein, e.g., between about 1 micrometer and 30 micrometers, or between about 1 and 25 micrometers. In some embodiments, the depth of the dummy TSV plate openings **222** may be between about 250 nanometers and 25 micrometers, including all values and ranges therein, e.g., between about 500 nanometers and 15 micrometers.

(29) The dummy TSV plate openings **222** are plate-like openings in that, for each of the dummy TSV plate openings **222**, the width may be at least 2 times smaller than each of the length and the depth, including all values and ranges therein, e.g., at least 3 times smaller or at least 5 times smaller. The length (a dimension measured along the y-axis of the example coordinate system shown in FIG. **2**) of the dummy TSV plate openings **222** may be comparable to the depth of these openings, or, in some embodiments, may be smaller than the depth of these openings. In some embodiments, the pitch of the dummy TSV plate openings **222** (i.e., center-to-center distance of adjacent openings **222**) may be between about 1.01 times and 2 times of the width of the dummy TSV plate openings **222**, including all values and ranges therein, e.g., between about 1.1 and 1.5 times of the width of the dummy TSV plate openings **222**. In some embodiments, a distance between the closest sidewalls of one of the TSV openings **220** and one of the dummy TSV plate openings **222** may be between about 100 nanometers and tens of micrometers, including all values and ranges therein, e.g., between about 500 nanometers and 50 micrometers.

(30) The method **100** may then proceed with a process **104** that includes conformally depositing a layer/liner of an insulator material into the openings formed in the process **102**. A result of this is illustrated with an IC structure **200**B, depicted in FIG. **2**B, showing a layer of an insulator material **204** deposited into the openings **220** and **222** formed in the process **102**. In some embodiments of the process **104**, a liner of the insulator material **204** may be deposited over sidewalls and bottoms of the TSV openings **220** and over sidewalls and bottoms of the dummy TSV plate openings **222** using any suitable techniques for conformally depositing materials onto selected surfaces, such as atomic layer deposition (ALD), chemical vapor deposition (CVD), plasma enhanced CVD (PECVD), or/and physical vapor deposition (PVD) processes such as sputter. In some embodiments, the insulator material **204** may include any suitable material for acting as an insulation barrier for the electrically conductive material that will later fill the TSV openings **220** and the dummy TSV plate openings **222**. Examples of such materials include, but are not limited to, silicon dioxide and silicon nitride. In some embodiments, the insulator material **204** may include any suitable material for acting as a capacitor insulator for future decoupling capacitors. Examples of such materials include, but are not limited to, dielectric materials known for their applicability in ICs, such as low-k dielectric materials. Examples of dielectric materials that may be used as the insulator material **204** may include, but are not limited to, silicon dioxide (SiO2), carbon-doped oxide (CDO), silicon nitride, fluorosilicate glass (FSG), silicon nitride, and organosilicates such as silsesquioxane, siloxane, or organosilicate glass. In some embodiments, the insulator material **204** may include organic polymers such as polyimide, polynorbornenes, benzocyclobutene, perfluorocyclobutane, or polytetrafluoroethylene (PTFE). Still other examples of low-k dielectric

materials that may be used as the insulator material **204** include silicon-based polymeric dielectrics such as hydrogen silsesquioxane (HSQ) and methylsilsesquioxane (MSQ). In some embodiments, a thickness of the insulator material **204** on the sidewalls and bottoms of the TSV openings **220** may be between about 1 and 7 nanometers, including all values and ranges therein, e.g., between about 2 and 5 nanometers.

(31) The method **100** may then proceed with a process **106** that includes conformally depositing a layer/liner of a barrier material into the openings formed in the process **102**. A result of this is illustrated with an IC structure **200**C, depicted in FIG. **2**C, showing a layer of a barrier material **206** deposited into the openings **220** and **222** formed in the process **102**. In particular, FIG. **2**C illustrates the embodiments where the barrier material **206** is deposited into the openings **220** and **222** that were first lined with the insulator material **204**. However, in other embodiments of the IC structure **200**, either the insulator material **204** or the barrier material **206** may be absent (i.e., in other embodiments of the method **100**, either the process **104** or the process **106** may be absent). In some embodiments of the process **106**, a liner of the barrier material **206** may be deposited over sidewalls and bottoms of the TSV openings **220** and over sidewalls and bottoms of the dummy TSV plate openings **222** using any suitable techniques for conformally depositing materials onto selected surfaces, such as ALD, CVD, PECVD, or/and PVD. In some embodiments, the barrier material **206** may include any suitable material for acting as a diffusion barrier for the electrically conductive material that will later fill the TSV openings **220** and the dummy TSV plate openings **222**. Examples of such materials include, but are not limited to, electrically conductive materials such as tantalum (Ta), tantalum nitride (TaN), titanium (Ti), titanium nitride (TiN), ruthenium (Ru), and cobalt (Co). In some embodiments, the liner of the barrier material **206** may help provide electrical conductivity, and, therefore, choice of materials used for the barrier material **206** may reflect that. For example, the barrier material **206** may include materials having a wide range of resistivities, such as in the range of 0.1 μΩ.Math.cm to 400 μΩ.Math.cm, including all values and ranges therein. More specifically than resistivity, in some implementations, choice of materials for the barrier material **206** may be made so that the resistance of the electrical connection satisfies appropriate requirements of the particular electrical circuit being implemented. Therefore, in some embodiments, high resistivity material(s) may still be used at an appropriately low thickness such that the net resistance satisfies circuit requirements. In some embodiments, a thickness of the barrier material **206** on the sidewalls and bottoms of the TSV openings **220** may be between about 5 and hundreds of nanometers, including all values and ranges therein, e.g., between about 10 and 200 nanometers.

(32) The method **100** may then continue with a process **108** that includes filling the TSV openings **220** and the dummy TSV plate openings **222** with an electrically conductive material. A result of this is illustrated with an IC structure **200**D, depicted in FIG. **2**D, showing an electrically conductive material **208** provided within the TSV openings **220** and the dummy TSV plate openings **222**, as well as over the upper surface of the IC structure **200**D. The electrically conductive material **208** may be deposited in the process **108** using a deposition technique such as, but not limited to, ALD, CVD, PECVD, PVD, or electroplating. The electrically conductive material **208** deposited in the process **108** may include one or more of any suitable electrically conductive materials (conductors). Such materials may include any suitable electrically conductive material, alloy, or a stack of multiple electrically conductive materials. In some embodiments, the electrically conductive material **208** may include one or more metals or metal alloys, with metals such as copper, ruthenium, palladium, platinum, cobalt, nickel, hafnium, zirconium, titanium, tantalum, and aluminum. In some embodiments, the electrically conductive material **208** may include one or more electrically conductive alloys, oxides (e.g., conductive metal oxides), carbides (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide, tungsten, tungsten carbide), or nitrides (e.g., hafnium nitride, zirconium nitride, titanium nitride, tantalum nitride, and aluminum nitride) of one or more metals.

(33) Next, the method **100** may include a process **110** that include removing the excess of the electrically conductive material **208** deposited in the process **108** over the upper surface of the IC structure (including over the TSV openings **220** and over the dummy TSV plate openings **222**) to expose the top (e.g., the second side **224-2**) of the support structure **202**. A result of this is illustrated with an IC structure **200**E, depicted in FIG. **2**E. A process of removing excess materials is typically referred to as “planarization.” In various embodiments, planarization of the process **110** may be performed using either wet or dry planarization processes. In one embodiment, planarization of the process **110** may be performed using chemical mechanical planarization (CMP), which may be understood as a process that utilizes a polishing surface, an abrasive and a slurry to remove the overburden of the electrically conductive material **208** and planarize the surface of the IC structure **200**D to expose the upper surface of the support structure **202** and the upper surface of the electrically conductive material **208** within the TSV openings **220** and within the dummy TSV plate openings **222**. Although FIG. **2**E and subsequent drawings illustrate that the TSV openings **220** and the dummy TSV plate openings **222** are fully filled with the electrically conductive material **208**, in other embodiments of the IC structure **200** the fill may be only partial in any of the TSV openings **220** and the dummy TSV plate openings **222** (e.g., the electrically conductive material **208** may only line the sidewalls and bottoms of any of the TSV openings **220** and the dummy TSV plate openings **222**).
(34) The TSV openings **220** at least partially filled with the electrically conductive material **208** form TSVs **230** (labeled in FIG. **2**E as TSVs **230-1** through **230-4**, corresponding to the TSV openings **220-1** through **220-4**). Similarly, the dummy TSV plate openings **222** at least partially filled with the electrically conductive material **208** form dummy TSV plates **232** (labeled in FIG. **2**E as dummy TSV plates **232-1** through **232-5**, corresponding to the dummy TSV plate openings **222-1** through **222-5**).
(35) Adjacent dummy TSV plates **232** may form first and second capacitor electrodes (in this case —capacitor plates) of decoupling capacitors of the IC structure **200**. To that end, the method **100** may also include a process **112** that includes providing contacts (i.e., electrical connections/interconnects) to the first and second capacitor plates of the decoupling capacitors formed by the dummy TSV plates **232**. A result of this is illustrated with an IC structure **200**F, depicted in FIGS. **2**F-**2**H, showing decoupling capacitors **234-1** (C1) through **234-4** (C4), and further showing TSV interconnects **240-1** and **240-2** for providing electrical connectivity to the electrically conductive material **208** of the TSVs **230**, and decoupling capacitor interconnects **242-1** and **242-2** for providing electrical connectivity to, respectively, first and second capacitor plates of the decoupling capacitors **234**.
(36) First and second capacitor electrodes of the decoupling capacitors **234**, formed by the adjacent dummy TSV plates **232**, are labeled in the IC structure **200**F as, respectively, capacitor electrodes E1.sub.Cx and E2.sub.Cx, where “E” stands for “electrode” (i.e., a capacitor electrode/plate) and “x” may be an integer indicating a given decoupling capacitor (e.g., x may be 1, 2, 3, or 4 to indicate one of the capacitors C1 through C4). For example, as labeled in the IC structure **200**F, the dummy TSV plates **232-1** and **232-2** may form, respectively, first and second capacitor electrodes E1.sub.C1 and E2.sub.C1 of the first decoupling capacitor **234-1**, the dummy TSV plates **232-3** and **232-2** may form, respectively, first and second capacitor electrodes E1.sub.C2 and E2.sub.C2 of the second decoupling capacitor **234-2**, the dummy TSV plates **232-3** and **232-4** may form, respectively, first and second capacitor electrodes E1.sub.C3 and E2.sub.C3 of the third decoupling capacitor **234-3**, and the dummy TSV plates **232-5** and **232-4** may form, respectively, first and second capacitor electrodes E1.sub.C4 and E2.sub.C4 of the fourth decoupling capacitor **234-4**. Thus, when multiple decoupling capacitors **234** are implemented, some of the dummy TSV plates **232** may serve as capacitor electrodes of different decoupling capacitors. For example, in the IC structure **200**F, each of the dummy TSV plates **232-2**, **232-3**, and **232-4** may be shared as a capacitor electrode of different decoupling capacitors **234**. For various decoupling capacitors **234**,

capacitor insulator (i.e., an electrically insulating material that separates first and second capacitor electrodes of a capacitor) may be a portion of the support structure **202** between the first and second capacitor electrodes and/or the insulator material **204** provided within the dummy TSV plate openings **222**.

(37) In some embodiments, the first capacitor electrodes E1.sub.Cx of various decoupling capacitors **234** may be coupled to the first decoupling capacitor interconnect **242-1**, while the second capacitor electrodes E2.sub.Cx of various decoupling capacitors **234** may be coupled to the second decoupling capacitor interconnect **242-2**, which may be electrically isolated from the first decoupling capacitor interconnect **242-1**. In some such embodiments, the first and second capacitor electrodes of multiple decoupling capacitors **234** may be provided in an alternating manner, as illustrated in the example of the IC structure **200**F. In some embodiments, the first capacitor electrodes E1.sub.Cx of various decoupling capacitors **234** may be coupled to a positive supply voltage (e.g., VDD), e.g., via the first decoupling capacitor interconnect **242-1**, while the second capacitor electrodes E2.sub.Cx of various decoupling capacitors **234** may be coupled to a negative supply voltage or a ground potential (e.g., VSS), e.g., via the second decoupling capacitor interconnect **242-2**. In other embodiments, this may be reversed, i.e., the first capacitor electrodes E1.sub.Cx of various decoupling capacitors **234** may be coupled to a negative supply voltage (e.g., VSS), e.g., via the first decoupling capacitor interconnect **242-1**, while the second capacitor electrodes E2.sub.Cx of various decoupling capacitors **234** may be coupled to a positive supply voltage or a ground potential (e.g., VDD), e.g., via the second decoupling capacitor interconnect **242-2**. In some embodiments, the first TSV interconnect **240-1** may be coupled to the electrically conductive material **208** of the TSVs **230-1** and **230-3**, while the second TSV interconnect **240-2** may be coupled to the electrically conductive material **208** of the TSVs **230-2** and **230-4**. In other embodiments, the number and the locations of any of the interconnects **240** and **242** may be different from what is shown in FIGS. **2**F-**2**H.

(38) The interconnects **240** and **242** may be formed of a contact material **210**, which may include any suitable electrically conductive material, such as any of the electrically conductive materials described above with reference to the electrically conductive material **208**, and may be provided within a layer of a dielectric material **212**, which may include any of the dielectric/interlayer dielectric (ILD) materials described above. The dielectric material **212** is not shown in the top-down view of FIG. **2**F in order to not obscure the decoupling capacitors **234**.

(39) The method **100** may conclude with a process **114** that includes thinning the back side of the support structure **202** to realize TSVs that extend between front and back sides of the support structure **202**. A result of this is illustrated with an IC structure **2001**, depicted in FIGS. **2**I-**2**J, showing that the back side **224-1** of the support structure **202** may be thinned until the electrically conductive material **208** of the TSVs **230** is exposed (i.e., is at the surface of the back side **224-1**) so that an electrical connection may be made to the TSVs **230**. The top-down view of the IC structure **2001** is not shown specifically because, in some embodiments, it may be substantially the same as the top-down view of the IC structure **200**F, shown in FIG. **2**F.

Example Devices

(40) The IC structures with decoupling capacitors based on dummy TSV plates disclosed herein may be included in any suitable electronic device. FIGS. **3-6** illustrate various examples of apparatuses that may include one or more of the IC structures disclosed herein.

(41) FIGS. **3**A and **3**B are top views of a wafer and dies that include one or more IC structures with one or more decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein. The wafer **1100** may be composed of semiconductor material and may include one or more dies **1102** having IC structures formed on a surface of the wafer **1100**. Each of the dies **1102** may be a repeating unit of a semiconductor product that includes any suitable IC structure (e.g., the IC structure **200** as shown in any of FIGS. **2**F-**2**I, or any further embodiments of the IC structure **200**). After the fabrication of the semiconductor product is complete (e.g., after

manufacture of one or more IC structures with one or more decoupling capacitors based on dummy TSV plates as described herein, included in a particular electronic component, e.g., in a transistor or in a memory device), the wafer **1100** may undergo a singulation process in which each of the dies **1102** is separated from one another to provide discrete "chips" of the semiconductor product. In particular, devices that include one or more IC structures with one or more decoupling capacitors based on dummy TSV plates as disclosed herein may take the form of the wafer **1100** (e.g., not singulated) or the form of the die **1102** (e.g., singulated). The die **1102** may include one or more transistors (e.g., one or more of the transistors **1240** of FIG. **4**, discussed below) and/or supporting circuitry to route electrical signals to the transistors, as well as any other IC components (e.g., one or more IC structures with one or more decoupling capacitors based on dummy TSV plates as discussed herein). In some embodiments, the wafer **1100** or the die **1102** may include a memory device (e.g., a static random-access memory (SRAM) device), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1102**. For example, a memory array formed by multiple memory devices may be formed on a same die **1102** as a processing device (e.g., the processing device **1402** of FIG. **6**) or other logic that is configured to store information in the memory devices or execute instructions stored in the memory array.

(42) FIG. **4** is a cross-sectional side view of an IC device **1200** that may include one or more IC structures with one or more decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein. The IC device **1200** may be formed on a substrate **1202** (e.g., the wafer **1100** of FIG. **3**A) and may be included in a die (e.g., the die **1102** of FIG. **3**B). The substrate **1202** may be any substrate as described herein. The substrate **1202** may be part of a singulated die (e.g., the dies **1102** of FIG. **3**B) or a wafer (e.g., the wafer **1100** of FIG. **3**A).

(43) The IC device **1200** may include one or more device layers **1204** disposed on the substrate **1202**. The device layer **1204** may include features of one or more transistors **1240** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the substrate **1202**. The device layer **1204** may include, for example, one or more source and/or drain (S/D) regions **1220**, a gate **1222** to control current flow in the transistors **1240** between the S/D regions **1220**, and one or more S/D contacts **1224** to route electrical signals to/from the S/D regions **1220**. The transistors **1240** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1240** are not limited to the type and configuration depicted in FIG. **4** and may include a wide variety of other types and configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

(44) Each transistor **1240** may include a gate **1222** formed of at least two layers, a gate electrode layer and a gate dielectric layer.

(45) The gate electrode layer may be formed on the gate interconnect support layer and may consist of at least one P-type workfunction metal or N-type workfunction metal, depending on whether the transistor is to be a PMOS or an NMOS transistor, respectively. In some implementations, the gate electrode layer may consist of a stack of two or more metal layers, where one or more metal layers are workfunction metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer or/and an adhesion layer.

(46) For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, and conductive metal oxides, e.g., ruthenium oxide. A P-type metal layer will enable the formation of a PMOS gate electrode with a workfunction that is between about 4.9 electron Volts (eV) and about 5.2 eV. For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, and carbides of these metals such as hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, aluminum carbide,

tungsten, tungsten carbide. An N-type metal layer will enable the formation of an NMOS gate electrode with a workfunction that is between about 3.9 eV and about 4.2 eV.

(47) In some embodiments, when viewed as a cross-section of the transistor **1240** along the source-channel-drain direction, the gate electrode may be formed as a U-shaped structure that includes a bottom portion substantially parallel to the surface of the substrate and two sidewall portions that are substantially perpendicular to the top surface of the substrate. In other embodiments, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the substrate and does not include sidewall portions substantially perpendicular to the top surface of the substrate. In other embodiments, the gate electrode may be implemented as a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may be implemented as one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers. In some embodiments, the gate electrode may consist of a V-shaped structure (e.g., when a fin of a FinFET transistor does not have a “flat” upper surface, but instead has a rounded peak).

(48) Generally, the gate dielectric layer of a transistor **1240** may include one layer or a stack of layers, and the one or more layers may include silicon oxide, silicon dioxide, and/or a high-k dielectric material. The high-k dielectric material included in the gate dielectric layer of the transistor **1240** may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric layer include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric layer to improve its quality when a high-k material is used.

(49) Although not specifically shown in FIG. **4**, the IC device **1200** may include one or more decoupling capacitors based on dummy TSV plates at any suitable location in the IC device **1200**.

(50) The S/D regions **1220** may be formed within the substrate **1202** adjacent to the gate **1222** of each transistor **1240**, using any suitable processes known in the art. For example, the S/D regions **1220** may be formed using either an implantation/diffusion process or a deposition process. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the substrate **1202** to form the S/D regions **1220**. An annealing process that activates the dopants and causes them to diffuse farther into the substrate **1202** may follow the ion implantation process. In the latter process, an epitaxial deposition process may provide material that is used to fabricate the S/D regions **1220**. In some implementations, the S/D regions **1220** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the S/D regions **1220** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1220**. In some embodiments, an etch process may be performed before the epitaxial deposition to create recesses in the substrate **1202** in which the material for the S/D regions **1220** is deposited.

(51) Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the transistors **1240** of the device layer **1204** through one or more interconnect layers disposed on the device layer **1204** (illustrated in FIG. **4** as interconnect layers **1206-1210**). For example, electrically conductive features of the device layer **1204** (e.g., the gate **1222** and the S/D contacts **1224**) may be electrically coupled with the interconnect structures **1228** of the interconnect layers **1206-1210**. The one or more interconnect layers **1206-1410** may form an ILD stack **1219** of the IC device **1200**.

(52) The interconnect structures **1228** may be arranged within the interconnect layers **1206-1210** to route electrical signals according to a wide variety of designs (in particular, the arrangement is not limited to the particular configuration of interconnect structures **1228** depicted in FIG. **4**). Although a particular number of interconnect layers **1206-1210** is depicted in FIG. **4**, embodiments of the present disclosure include IC devices having more or fewer interconnect layers than depicted.
(53) In some embodiments, the interconnect structures **1228** may include trench contact structures **1228***a* (sometimes referred to as "lines") and/or via structures **1228***b* (sometimes referred to as "holes") filled with an electrically conductive material such as a metal. The trench contact structures **1228***a* may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the substrate **1202** upon which the device layer **1204** is formed. For example, the trench contact structures **1228***a* may route electrical signals in a direction in and out of the page from the perspective of FIG. **4**. The via structures **1228***b* may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the substrate **1202** upon which the device layer **1204** is formed. In some embodiments, the via structures **1228***b* may electrically couple trench contact structures **1228***a* of different interconnect layers **1206-1210** together.
(54) The interconnect layers **1206-1210** may include a dielectric material **1226** disposed between the interconnect structures **1228**, as shown in FIG. **4**. The dielectric material **1226** may take the form of any of the embodiments of the dielectric material provided between the interconnects of the IC structures disclosed herein, for example any of the embodiments discussed herein with reference to the dielectric material **204** or **212**, described herein.
(55) In some embodiments, the dielectric material **1226** disposed between the interconnect structures **1228** in different ones of the interconnect layers **1206-1210** may have different compositions. In other embodiments, the composition of the dielectric material **1226** between different interconnect layers **1206-1210** may be the same.
(56) A first interconnect layer **1206** (referred to as Metal 1 or "M1") may be formed directly on the device layer **1204**. In some embodiments, the first interconnect layer **1206** may include trench contact structures **1228***a* and/or via structures **1228***b*, as shown. The trench contact structures **1228***a* of the first interconnect layer **1206** may be coupled with contacts (e.g., the S/D contacts **1224**) of the device layer **1204**.
(57) A second interconnect layer **1208** (referred to as Metal 2 or "M2") may be formed directly on the first interconnect layer **1206**. In some embodiments, the second interconnect layer **1208** may include via structures **1228***b* to couple the trench contact structures **1228***a* of the second interconnect layer **1208** with the trench contact structures **1228***a* of the first interconnect layer **1206**. Although the trench contact structures **1228***a* and the via structures **1228***b* are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **1208**) for the sake of clarity, the trench contact structures **1228***a* and the via structures **1228***b* may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some embodiments.
(58) A third interconnect layer **1210** (referred to as Metal 3 or "M3") (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1208** according to similar techniques and configurations described in connection with the second interconnect layer **1208** or the first interconnect layer **1206**.
(59) The IC device **1200** may include a solder resist material **1234** (e.g., polyimide or similar material) and one or more bond pads **1236** formed on the interconnect layers **1206-1210**. The bond pads **1236** may be electrically coupled with the interconnect structures **1228** and configured to route the electrical signals of the transistor(s) **1240** to other external devices. For example, solder bonds may be formed on the one or more bond pads **1236** to mechanically and/or electrically couple a chip including the IC device **1200** with another component (e.g., a circuit board). The IC device **1200** may have other alternative configurations to route the electrical signals from the

interconnect layers **1206-1210** than depicted in other embodiments. For example, the bond pads **1236** may be replaced by or may further include other analogous features (e.g., posts) that route the electrical signals to external components.

(60) FIG. **5** is a cross-sectional side view of an IC device assembly **1300** that may include components having or being associated with (e.g., being electrically connected by means of) one or more IC structures with decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein. The IC device assembly **1300** includes a number of components disposed on a circuit board **1302** (which may be, e.g., a motherboard). The IC device assembly **1300** includes components disposed on a first face **1340** of the circuit board **1302** and an opposing second face **1342** of the circuit board **1302**; generally, components may be disposed on one or both faces **1340** and **1342**. In particular, any suitable ones of the components of the IC device assembly **1300** may include any of the decoupling capacitors based on dummy TSV plates, disclosed herein.

(61) In some embodiments, the circuit board **1302** may be a printed circuit board (PCB) including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1302**. In other embodiments, the circuit board **1302** may be a non-PCB substrate.

(62) The IC device assembly **1300** illustrated in FIG. **5** includes a package-on-interposer structure **1336** coupled to the first face **1340** of the circuit board **1302** by coupling components **1316**. The coupling components **1316** may electrically and mechanically couple the package-on-interposer structure **1336** to the circuit board **1302** and may include solder balls (as shown in FIG. **5**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

(63) The package-on-interposer structure **1336** may include an IC package **1320** coupled to an interposer **1304** by coupling components **1318**. The coupling components **1318** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1316**. Although a single IC package **1320** is shown in FIG. **5**, multiple IC packages may be coupled to the interposer **1304**; indeed, additional interposers may be coupled to the interposer **1304**. The interposer **1304** may provide an intervening substrate used to bridge the circuit board **1302** and the IC package **1320**. The IC package **1320** may be or include, for example, a die (the die **1102** of FIG. 3B), an IC device (e.g., the IC device **1200** of FIG. **4**), or any other suitable component. In some embodiments, the IC package **1320** may include one or more decoupling capacitors based on dummy TSV plates, as described herein. Generally, the interposer **1304** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **1304** may couple the IC package **1320** (e.g., a die) to a ball grid array (BGA) of the coupling components **1316** for coupling to the circuit board **1302**. In the embodiment illustrated in FIG. **5**, the IC package **1320** and the circuit board **1302** are attached to opposing sides of the interposer **1304**; in other embodiments, the IC package **1320** and the circuit board **1302** may be attached to a same side of the interposer **1304**. In some embodiments, three or more components may be interconnected by way of the interposer **1304**.

(64) The interposer **1304** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, a ceramic material, or a polymer material such as polyimide. In some implementations, the interposer **1304** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1304** may include metal interconnects **1308** and vias **1310**, including but not limited to TSVs **1306**. The interposer **1304** may further include embedded devices **1314**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers,

sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency (RF) devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1304**. The interposer **1304** may further include one or more decoupling capacitors based on dummy TSV plates, as described herein. The package-on-interposer structure **1336** may take the form of any of the package-on-interposer structures known in the art.

(65) The IC device assembly **1300** may include an IC package **1324** coupled to the first face **1340** of the circuit board **1302** by coupling components **1322**. The coupling components **1322** may take the form of any of the embodiments discussed above with reference to the coupling components **1316**, and the IC package **1324** may take the form of any of the embodiments discussed above with reference to the IC package **1320**.

(66) The IC device assembly **1300** illustrated in FIG. **5** includes a package-on-package structure **1334** coupled to the second face **1342** of the circuit board **1302** by coupling components **1328**. The package-on-package structure **1334** may include an IC package **1326** and an IC package **1332** coupled together by coupling components **1330** such that the IC package **1326** is disposed between the circuit board **1302** and the IC package **1332**. The coupling components **1328** and **1330** may take the form of any of the embodiments of the coupling components **1316** discussed above, and the IC packages **1326** and **1332** may take the form of any of the embodiments of the IC package **1320** discussed above. The package-on-package structure **1334** may be configured in accordance with any of the package-on-package structures known in the art.

(67) FIG. **6** is a block diagram of an example computing device **1400** that may include one or more components including one or more IC structures with one or more decoupling capacitors based on dummy TSV plates in accordance with any of the embodiments disclosed herein. For example, any suitable ones of the components of the computing device **1400** may include a die (e.g., the die **1102** of FIG. **3**B) having one or more decoupling capacitors based on dummy TSV plates as described herein. Any one or more of the components of the computing device **1400** may include, or be included in, an IC device **1200** (FIG. **4**). Any one or more of the components of the computing device **1400** may include, or be included in, an IC device assembly **1300** (FIG. **5**).

(68) A number of components are illustrated in FIG. **6** as included in the computing device **1400**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the computing device **1400** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

(69) Additionally, in various embodiments, the computing device **1400** may not include one or more of the components illustrated in FIG. **6**, but the computing device **1400** may include interface circuitry for coupling to the one or more components. For example, the computing device **1400** may not include a display device **1406**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **1406** may be coupled. In another set of examples, the computing device **1400** may not include an audio input device **1424** or an audio output device **1408** but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1424** or audio output device **1408** may be coupled.

(70) The computing device **1400** may include a processing device **1402** (e.g., one or more processing devices). As used herein, the term "processing device" or "processor" may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **1402** may include one or more digital signal processors (DSPs), application-specific ICs (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The computing device **1400**

may include a memory **1404**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random-access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **1404** may include memory that shares a die with the processing device **1402**. This memory may be used as cache memory and may include embedded dynamic random-access memory (eDRAM) or spin transfer torque magnetic random-access memory (STT-MRAM).

(71) In some embodiments, the computing device **1400** may include a communication chip **1412** (e.g., one or more communication chips). For example, the communication chip **1412** may be configured for managing wireless communications for the transfer of data to and from the computing device **1400**. The term "wireless" and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

(72) The communication chip **1412** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultramobile broadband (UMB) project (also referred to as "3GPP2"), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1412** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High-Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1412** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1412** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1412** may operate in accordance with other wireless protocols in other embodiments. The computing device **1400** may include an antenna **1422** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

(73) In some embodiments, the communication chip **1412** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1412** may include multiple communication chips. For instance, a first communication chip **1412** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1412** may be dedicated to longer-range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **1412** may be dedicated to wireless communications, and a second communication chip **1412** may be dedicated to wired communications.

(74) The computing device **1400** may include battery/power circuitry **1414**. The battery/power circuitry **1414** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the computing device **1400** to an energy source separate from the computing device **1400** (e.g., AC line power).

(75) The computing device **1400** may include a display device **1406** (or corresponding interface circuitry, as discussed above). The display device **1406** may include any visual indicators, such as a

heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display, for example.

(76) The computing device **1400** may include an audio output device **1408** (or corresponding interface circuitry, as discussed above). The audio output device **1408** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds, for example.

(77) The computing device **1400** may include an audio input device **1424** (or corresponding interface circuitry, as discussed above). The audio input device **1424** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

(78) The computing device **1400** may include a global positioning system (GPS) device **1418** (or corresponding interface circuitry, as discussed above). The GPS device **1418** may be in communication with a satellite-based system and may receive a location of the computing device **1400**, as known in the art.

(79) The computing device **1400** may include an other output device **1410** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1410** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

(80) The computing device **1400** may include an other input device **1420** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1420** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

(81) The computing device **1400** may have any desired form factor, such as a hand-held or mobile computing device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultramobile personal computer, etc.), a desktop computing device, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable computing device. In some embodiments, the computing device **1400** may be any other electronic device that processes data.

SELECT EXAMPLES

(82) The following paragraphs provide various examples of the embodiments disclosed herein. Example 1 provides an IC structure that includes a support structure (e.g., a substrate), having a first side and a second side, opposite the first side; a TSV, extending between the first side and the second side; and a decoupling capacitor (C1) having a first capacitor plate (E1.sub.C1), a second capacitor plate (E2.sub.C1), and a capacitor insulator (silicon has dielectric constant of .sup.˜12, would contribute to capacitance) between the first capacitor plate and the second capacitor plate. In such an IC structure, the first capacitor plate is a first opening in the support structure, the first opening at least partially filled with a first electrically conductive material, while the second capacitor plate is a second opening in the support structure, the second opening at least partially filled with a second electrically conductive material (which may be the same as or different from the first electrically conductive material). Each of the first opening and the second opening extends from the second side towards, but not reaching, the first side of the support structure. Each of the first opening and the second opening has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth, the width is a dimension of the first or second opening in a direction of an x-axis of an x-y-z coordinate system, the length is a dimension of the first or second opening in a direction of a y-axis of the x-y-z coordinate system, and the depth is a dimension of the first or second opening in a direction of a z-axis of the x-y-z coordinate system, each of the x, y, and z-axis being perpendicular to the other two of the axis.

(83) Example 2 provides the IC structure according to example 1, where the capacitor insulator

includes a portion of the support structure between the first capacitor plate and the second capacitor plate.

(84) Example 3 provides the IC structure according to examples 1 or 2, where at least one of the first opening and the second opening includes a liner of an insulator material (e.g., material **204** shown in the present drawings) on sidewalls and a bottom of the opening.

(85) Example 4 provides the IC structure according to example 3, where the capacitor insulator includes at least a portion of the liner of the insulator material.

(86) Example 5 provides the IC structure according to examples 3 or 4, where a thickness of the liner of the insulator material is between about 1 and 7 nanometers.

(87) Example 6 provides the IC structure according to any one of examples 3-5, further including a liner of the insulator material on sidewalls and a bottom of the TSV.

(88) Example 7 provides the IC structure according to any one of examples 3-6, where the insulator material is silicon oxide or silicon nitride.

(89) Example 8 provides the IC structure according to any one of the preceding examples, where at least one of the first opening and the second opening includes a liner of a barrier material on sidewalls and a bottom of the opening.

(90) Example 9 provides the IC structure according to any one of the preceding examples, where, at least for one of the first opening and the second opening, the depth is between about 1 and 25 micrometers.

(91) Example 10 provides the IC structure according to any one of the preceding examples, where, at least for one of the first opening and the second opening, the depth is between about 1.1 and 4 times smaller than a distance between the first side and the second side of the support structure.

(92) Example 11 provides the IC structure according to any one of the preceding examples, where, at least for one of the first opening and the second opening, the width is between about 250 and 5000 nanometers.

(93) Example 12 provides the IC structure according to any one of the preceding examples, where the width of each of the first opening and the second opening is smaller than a width of the TSV (i.e., a dimension of the TSV in an x-y plane, e.g., in a direction of an x-axis, of the x-y-z coordinate system shown in the present drawings), e.g., at least 2 times smaller or at least 4 times smaller.

(94) Example 13 provides the IC structure according to any one of the preceding examples, where the first capacitor plate is coupled to a positive supply voltage and the second capacitor plate is coupled to a negative supply voltage or a ground voltage.

(95) Example 14 provides the IC structure according to any one of the preceding examples, where the decoupling capacitor is a first decoupling capacitor (C1), the IC structure further includes a second decoupling capacitor (C2) having a first capacitor plate, a second capacitor plate, and a capacitor insulator between the first capacitor plate and the second capacitor plate, the second capacitor plate of the second decoupling capacitor (E2.sub.C2) is the second capacitor plate of the first decoupling capacitor (E2.sub.C1), the first capacitor plate of the second decoupling capacitor (E1.sub.C2) is a third opening in the support structure, the third opening at least partially filled with a third electrically conductive material (which may be the same as or different from the first and second electrically conductive materials), the second opening is between the first opening and the third opening, and the third opening extends from the second side towards, but not reaching, the first side of the support structure. The third opening has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth, the width is a dimension of the opening in a direction of an x-axis of an x-y-z coordinate system, the length is a dimension of the opening in a direction of a y-axis of the x-y-z coordinate system, and the depth is a dimension of the opening in a direction of a z-axis of the x-y-z coordinate system, each of the x, y, and z-axis being perpendicular to the other two of the axis.

(96) Example 15 provides the IC structure according to example 14, where the IC structure further

includes a third decoupling capacitor (C3) and a fourth decoupling capacitor (C4), each having a first capacitor plate, a second capacitor plate, and a capacitor insulator between the first capacitor plate and the second capacitor plate. The first capacitor plate of the third decoupling capacitor (E1.sub.C3) is the first capacitor plate of the second decoupling capacitor (E1.sub.C2). The second capacitor plate of the third decoupling capacitor (E2.sub.C3) is the second capacitor plate of the fourth decoupling capacitor (E2.sub.C4). The second capacitor plate of the fourth decoupling capacitor (E2.sub.C4) is a fourth opening in the support structure, the fourth opening at least partially filled with a fourth electrically conductive material (which may be the same as or different from any of the first, second, and third electrically conductive materials). The first capacitor plate of the fourth decoupling capacitor (E1.sub.C4) is a fifth opening in the support structure, the fifth opening at least partially filled with a fifth electrically conductive material (which may be the same as or different from any of the first, second, third, and fourth electrically conductive materials). The fourth opening is between the third opening and the fifth opening. Each of the fourth opening and the fifth opening extends from the second side towards, but not reaching, the first side of the support structure, and each of the fourth opening and the fifth opening has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth, the width is a dimension of the opening in a direction of an x-axis of an x-y-z coordinate system, the length is a dimension of the opening in a direction of a y-axis of the x-y-z coordinate system, and the depth is a dimension of the opening in a direction of a z-axis of the x-y-z coordinate system, each of the x, y, and z-axis being perpendicular to the other two of the axis.

(97) Example 16 provides the IC structure according to example 15, where, for each of the first decoupling capacitor, the second decoupling capacitor, the third decoupling capacitor, and the fourth decoupling capacitor, the first capacitor plate is coupled to a positive supply voltage and the second capacitor plate is coupled to a negative supply voltage or a ground voltage.

(98) Example 17 provides an IC package that includes an IC die having a first side and a second side, opposite the first side; and a further IC component, coupled to the IC die. The IC die has a first side and a second side, opposite the first side, the IC die includes a decoupling capacitor (C1) having a first capacitor plate (E1.sub.C1), a second capacitor plate (E2.sub.C1), and a capacitor insulator between the first capacitor plate and the second capacitor plate, the first capacitor plate is a first opening in the support structure, the first opening at least partially filled with a first electrically conductive material, and the second capacitor plate is a second opening in the support structure, the second opening at least partially filled with a second electrically conductive material (which may be the same as or different from the first electrically conductive material). Furthermore, each of the first opening and the second opening extends from the second side towards, but not reaching, the first side of the support structure, each of the first opening and the second opening has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth, the width is a dimension of the first or second opening in a direction of an x-axis of an x-y-z coordinate system, the length is a dimension of the first or second opening in a direction of a y-axis of the x-y-z coordinate system, and the depth is a dimension of the first or second opening in a direction of a z-axis of the x-y-z coordinate system, each of the x, y, and z-axis being perpendicular to the other two of the axis.

(99) Example 18 provides the IC package according to example 17, where the IC die further includes a TSV extends between the first side and the second side of the IC die.

(100) Example 19 provides the IC package according to examples 17 or 18, where, for each of the first opening and the second opening, the depth is between about 1.1 and 4 times smaller than a distance between the first side and the second side of the IC die.

(101) Example 20 provides the IC package according to any one of examples 17-19, where the further component is one of a package substrate, a flexible substrate, or an interposer.

(102) Example 21 provides the IC package according to any one of examples 17-20, where the further component is coupled to the IC die via one or more first level interconnects.

(103) Example 22 provides the IC package according to example 21, where the one or more first level interconnects include one or more solder bumps, solder posts, or bond wires.
(104) Example 23 provides a method for fabricating an IC structure. The method includes providing, in a support structure, an opening for a through-silicon via (TSV), and a first and a second openings for a decoupling capacitor, where each of the first opening and the second opening is a blind opening in the support structure, each of the first opening and the second opening has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth, the width is a dimension of the first or second opening in a direction of an x-axis of an x-y-z coordinate system, the length is a dimension of the first or second opening in a direction of a y-axis of the x-y-z coordinate system, and the depth is a dimension of the first or second opening in a direction of a z-axis of the x-y-z coordinate system, each of the x, y, and z-axis being perpendicular to the other two of the axis, and the depth of the first and second openings for the decoupling capacitor is between about 1.1 and 4 times smaller than a depth of the opening for the TSV. The method also includes providing a first capacitor plate of the decoupling capacitor by at least partially filling the first opening with a first electrically conductive material; and providing a second capacitor plate of the decoupling capacitor by at least partially filling the second opening with a second electrically conductive material (which may be the same as or different from the first electrically conductive material).
(105) Example 24 provides the method according to example 23, further including providing a liner of an insulator material (e.g., material **204** shown in the present drawings) on sidewalls and a bottom of at least one of the first opening and the second opening before providing the first capacitor plate and the second capacitor plate.
(106) Example 25 provides the method according to examples 23 or 24, further including processes for fabricating the IC structure according to any one of the preceding examples (e.g., the IC structure according to any one of examples 1-16) and/or processes for fabricating the IC package according to any one of the preceding examples (e.g., the IC package according to any one of examples 17-22).
(107) The above description of illustrated implementations of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. These modifications may be made to the disclosure in light of the above detailed description.

Claims

1. An integrated circuit (IC) structure, comprising: a support having a first side and a second side, opposite the first side; a via extending between the first side and the second side of the support, wherein the via has a first end and a second end, the second end is opposite the first end, a conductive material of the via at the first end is at the first side of the support, and the conductive material of the via at the second end is at the second side of the support; and a capacitor having a first capacitor plate, a second capacitor plate, and a capacitor insulator between the first capacitor plate and the second capacitor plate, wherein: the first capacitor plate and the second capacitor plate are opposing plates of the capacitor, the first capacitor plate includes a first structure of a first conductive material, the first structure extending from the second side towards the first side of the support, the second capacitor plate includes a second structure of a second conductive material, the second structure extending from the second side towards the first side of the support, the first capacitor plate and the second capacitor plate are substantially parallel to one another, the conductive material of the via is electrically isolated from the first capacitor plate and from the second capacitor plate, and a width of the via is greater than a width of the first capacitor plate and

a width of the second capacitor plate.
2. The IC structure according to claim 1, wherein the capacitor insulator includes a portion of the support between the first capacitor plate and the second capacitor plate.
3. The IC structure according to claim 1, wherein the first capacitor plate includes a liner of an insulator material on sidewalls and a bottom of the first structure of the first conductive material.
4. The IC structure according to claim 3, wherein the insulator material includes silicon and oxygen or nitrogen.
5. The IC structure according to claim 1, wherein the first capacitor plate includes a liner of a barrier material on sidewalls and a bottom of the first structure of the first conductive material.
6. The IC structure according to claim 1, wherein, the first structure of the first conductive material extends from the second side towards the first side of the support by a depth between about 1 micrometer and 25 micrometers.
7. The IC structure according to claim 1, wherein the first structure of the first conductive material extends from the second side towards the first side of the support by a depth, and the depth is between about 1.1 and 4 times smaller than a distance between the first side and the second side.
8. The IC structure according to claim 1, further comprising: a first contact to couple the first capacitor plate to a first voltage; and a second contact to couple the second capacitor plate to a second voltage.
9. The IC structure according to claim 1, wherein: the capacitor is a first capacitor, the IC structure further includes a second capacitor having a first capacitor plate, a second capacitor plate, and a capacitor insulator between the first capacitor plate and the second capacitor plate, the second capacitor plate of the second capacitor is the second capacitor plate of the first capacitor, the first capacitor plate of the second capacitor is a third structure of a third conductive material, the third structure extending from the second side towards, but not reaching, the first side of the support, the second capacitor plate is between the first capacitor plate and the third structure, the third structure and the second structure are substantially parallel to one another, and the conductive material of the via is electrically isolated from the first capacitor plate of the second capacitor.
10. The IC structure according to claim 9, wherein: the IC structure further includes a third capacitor and a fourth capacitor, each having a first capacitor plate, a second capacitor plate, and a capacitor insulator between the first capacitor plate and the second capacitor plate, the first capacitor plate of the third capacitor is the first capacitor plate of the second capacitor, the second capacitor plate of the third capacitor is the second capacitor plate of the fourth capacitor, the second capacitor plate of the fourth capacitor is a fourth structure of a fourth conductive material, the fourth structure extending from the second side towards, but not reaching, the first side of the support, the first capacitor plate of the fourth capacitor is a fifth structure of a fifth conductive material, the fifth structure extending from the second side towards, but not reaching, the first side of the support, the fourth structure is between the third structure and the fifth structure, and the third structure, the fourth structure, and the fifth structure are substantially parallel to one another.
11. The IC structure according to claim 10, wherein the conductive material of the via is electrically isolated from the fourth structure and the fifth structure.
12. The IC structure according to claim 1, wherein: each of the first structure and the second structure has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth.
13. The IC structure according to claim 1, wherein the first structure and the second structure do not reach the first side of the support.
14. The IC structure according to claim 1, wherein the first structure and the second structure are planar structures.
15. The IC structure according to claim 1, wherein the first capacitor plate includes a liner of an insulator material on sidewalls and a bottom of the first structure of the first conductive material, and wherein the capacitor insulator includes at least a portion of the liner of the insulator material.

16. An integrated circuit (IC) package, comprising: an IC die; and an IC component, coupled to the IC die, wherein: the IC die has a first side and a second side, opposite the first side, the IC die includes a through via that extends between the first side of the IC die and the second side of the IC die and has a first end at the first side of the IC die and a second end at the second side of the IC die, the IC die further includes a capacitor having a first capacitor plate, a second capacitor plate, and a capacitor insulator between the first capacitor plate and the second capacitor plate, the first capacitor plate includes a first planar structure of a first conductive material, the first planar structure extending from the second side towards, but not reaching, the first side of the IC die, the second capacitor plate includes a second planar structure of a second conductive material, the second planar structure extending from the second side towards, but not reaching, the first side of the IC die, the first capacitor plate and the second capacitor plate are substantially parallel to one another, a conductive material of the through via is electrically isolated from the first capacitor plate and from the second capacitor plate, and a width of the through via is greater than a width of the first capacitor plate and a width of the second capacitor plate.
17. The IC package according to claim 16, wherein the width of the through via is between 1.5 and 10 times greater than the width of the first capacitor plate and the width of the second capacitor plate.
18. The IC package according to claim 16, wherein: the first planar structure and the second planar structure extend from the second side towards, but not reaching, the first side of the IC die, and each of the first planar structure and the second planar structure has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth.
19. The IC package according to claim 16, wherein the IC component is one of another IC die, a package substrate, a circuit board, or an interposer.
20. A method for fabricating an integrated circuit (IC) structure, the method comprising: providing a via extending from a first side to a second side of a support, wherein the second side is opposite the first side, the via has a first end and a second end, the second end is opposite the first end, a conductive material of the via at the first end is at the first side of the support, and the conductive material of the via at the second end is at the second side of the support; and providing a capacitor having a first capacitor plate, a second capacitor plate, and a capacitor insulator between the first capacitor plate and the second capacitor plate, wherein: the first capacitor plate and the second capacitor plate are opposing plates of the capacitor, the first capacitor plate includes a first structure of a first conductive material, the first structure extending from the second side towards the first side of the support, the second capacitor plate includes a second structure of a second conductive material, the second structure extending from the second side towards the first side of the support, the first capacitor plate and the second capacitor plate are substantially parallel to one another, the conductive material of the via is electrically isolated from the first capacitor plate and from the second capacitor plate, and a width of the via is greater than a width of the first capacitor plate and a width of the second capacitor plate.
21. The method according to claim 19, wherein: each of the first structure and the second structure has a width, a length, and a depth such that the width is at least about 2 times smaller than each of the length and the depth, and the first structure extends from the second side towards the first side of the support by a distance that is between about 1.1 and 4 times smaller than a distance between the first side and the second side.